Ayush Anand

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EXPERIENCE

Software Engineer — Gurugram, India
Zomato Limited — Dec 2024 - Present

- Built an end-to-end automated outbound calling platform that enabled sales/marketing to target churned merchants; integrated telecom APIs with an AI voice bot to conduct feedback calls at 300 calls/min, transcribed and summarized conversations, and structured insights into actionable buckets—driving 60% higher merchant engagement.
- Designed and implemented a high-volume vendor ledger and payout system to accrue invoices, reconcile against debit/credit notes, and enable daily finance approvals; processed 10K+ vendor disbursements via SFTP bank integrations with event-driven workflows (Go, Kafka, DynamoDB, Redis), eliminating 20 hours/week of manual reconciliation and reducing payout errors.
- Built a scalable vendor invoice ingestion and approval system that digitized invoices from existing workflows using in-house LLM services, validated extracted data against POs with rule-based checks, routed conflicts for human approval/rejection, and auto-posted approved invoices into the ledger—reducing processing costs 10x and cutting turnaround time by 50%.
- Increased completion of critical SalesOps tasks by 35% by shipping a real-time push notification platform (Go + Kafka); designed idempotent consumers, retry/backoff, and rules-based routing so the right teams acted on the right events fast.
- Strengthened reliability and security across a microservices stack by optimizing high-latency APIs (profiling + otel metrics) and remediating critical vulnerabilities in old systems;
- Built an integrated computer vision pipeline that ingested live CCTV streams, detected SOP breaches using vision models, and triggered real-time alerts under 300ms latency, reducing warehouse pilferage costs by 30% within the first week.

Software Engineer — Remote, UK
DB News (Early-stage fintech startup building real-time financial news infrastructure) — Nov 2023 - Nov 2024

- Designed and built a high-performance news streaming platform that ingested critical market events and broadcasted filtered, real-time updates to subscribed users under 300ms using a WebSocket-as-a-Service and Pub/Sub architecture—driving.
- Optimized Postgres database operations to achieve a remarkable 3ms latency for reads/writes through indexing and partitioning techniques, optimizing overall system responsiveness.
- Created robust user session management using Redis, enabling seamless authentication and authorization processes while managing over 3,000 daily write operations and 15,000 daily read operations on the database.

Software Engineer Intern — Mumbai, India
JP Morgan Chase & Co — May 2024 - July 2024

- Achieved a 70% reduction in time-to-recovery from failed Risk and PnL jobs by developing a Flask-based API utility with thread pooling, local caching, and RQL for JP Morgan's in-house trading platform.
- Accelerated system performance and data retrieval by utilizing an in-house distributed object store with indexing, resulting in a 100% increase in meeting End-of-Day SLAs.
- Strengthened overall system efficiency and reliability by implementing load balancing and optimized data processing frameworks to deliver high-impact business solutions.

Software Engineer Intern — Remote
Google Summer of Code — May 2022 - Sept 2022

- Architected and optimized a Python package to improve performance and scalability for marine researchers, reducing data processing time by over 70%. *github.com/iobis/pyobis*.
- Refactored codebase to reduce size by over 50%, improving efficiency and enabling widespread use by researchers at the University of South Florida.
- Implemented 5+ new features, resolved 27+ open issues, and boosted performance by over 75% within 4 weeks.

SKILLS

Programming Languages	Golang, Python, C++, C#, TypeScript, JavaScript, Postgres, MySQL, React, Redis
Libraries/Tools	gRPC, Git, Node.js, CI/CD, Apache Kafka, AWS, Docker, REST API, Kuma distributed mesh

EDUCATION

National Institute of Technology — Durgapur, India
Bachelor of Technology in Computer Science and Engineering — GPA: 9.46/10

- *Courses: Data Structure and Algorithms, Operating Systems, Database Management System, Computer Networks*
- Organisations: Vice Chairperson @ Indian Society for Technical Education (ISTE) Students' Chapter

PROJECTS

crazyhttp - An modern async HTTP/3 (HTTP over QUIC) Server Library - [on Github]
Technologies used: Golang, QUIC, BoringSSL, IP, Networks, Linux
1. Developed an asynchronous HTTP/3 (HTTP over QUIC) [H/1.1 + H/2 compliant] library for server creation, integrating event loops (libev) and native Linux primitives, while leveraging Cloudflare's quiche implementation.
2. Achieved nearly 153x faster startup times and reduced p95 RTT by 50.97% compared to FastAPI.

crazygraphstore - An LSM Tree based fast storage engine for graph databases - [on Github]
Technologies used: Golang, Linux Primitives
1. Built a storage engine for graph databases from scratch in Go, designed for speed and high write throughput, beating Neo4j in benchmarks by 23% in write throughput. Benchmarked upto 500K+ reads/sec and 50K+ writes/sec.
2. Has a 3-phase write pipeline (memory → disk → compaction) with crash recovery and bloom-filter lookups on SSTables — eliminating redundant disk I/O and keeping reads fast at high throughput.

ACHIEVEMENTS

- Knight on LeetCode - Rating: 1872 - Top 5.22%
- Secured first rank among 850+ students with a 9.9/10 SGPA, demonstrating academic excellence.
- Open Source Contributor to JPMorgan/Salt DS and Todoist/Typist, maintainer of two research python packages.